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Emotion–Behavior Decoupling in Individuals With Schizophrenia, Bipolar Disorder, and Major Depressive Disorder

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Failure in translating emotional salience into effortful behavior is thought to be a core feature of anhedonia and avolition in individuals with schizophrenia (SCZ), but little is known about emotion-behavior coupling in individuals with bipolar disorder (BD) and major depressive disorder (MDD). In this study, we compared emotion–behavior correspondence in participants with SCZ, BD, and MDD. Forty-two participants with SCZ, 44 participants with MDD, 43 participants with BD, and 43 healthy controls were recruited. A computerized anticipatory and consummatory pleasure task was used to evaluate emotion–behavior correspondence. Clinical ratings of negative symptoms and self-report anhedonia questionnaires were also administered. We found that participants with SCZ, MDD, and BD exhibited different levels of negative symptoms and self-reported anhedonia, as well as emotion–behavior decoupling. In SCZ participants, both desirable and undesirable images elicited lower correspondence between self-reported liking and behavior. In MDD and BD participants, undesirable images elicited lower emotion–behavior correspondence under both direct stimulus presentation and representation conditions, whereas deficits in emotion–behavior coupling under desirable conditions were only observed when stimuli were present. Taken together, emotion–behavior decoupling showed both common and unique patterns in participants with SCZ, MDD, and BD, and showed some associations with negative symptoms and anhedonia across the combined clinical sample. This finding may be helpful for early identification and the development of novel interventions for different psychiatric diagnoses.

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General Scientific Summary
Failure to maintain motivation to achieve key goals is a core feature shared by patients with schizophrenia, bipolar disorder, and major depressive disorder. However, it is unclear whether such a failure in translating emotion into effortful behavior in these three clinical groups follows the same pattern. The authors administered a computer-based test to examine the pattern of dissociation between emotion and effortful behavior in these three types of patients and a group of healthy volunteers. They found that although all three patient groups exhibited a lowered capacity to experience pleasure as well as a lack of motivation, the pattern of emotion–behavior dissociation is different across schizophrenia, bipolar disorder, and major depressive disorder.

Keywords: anhedonia, transdiagnostic, emotion–behavior decoupling, mental disorders

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Anhedonia and *avolition* refer to the diminished ability to experience pleasure and to expend effortful behavior respectively. These two symptoms exist across different diagnostic entities, including schizophrenia (SCZ), bipolar disorder (BD), and major depressive disorder (MDD; Treadway & Zald, 2013), and contribute to poor treatment response and impaired social functioning (Spijker, Bijl, de Graaf, & Nolen, 2001). Evidence from affective neuroscience suggests that anhedonia can be parsed into consummatory anhedonia (affecting the "liking" component) and anticipatory anhedonia (affecting the "wanting" component; Berridge & Kringelbach, 2015).

Several previous studies have reported that individuals with first-episode and chronic SCZ exhibit an intact ability to enjoy positively valenced stimuli ("liking"), but lack the motivation to pursue positive outcomes ("wanting"; Cohen & Minor, 2010; Llerena, Strauss, & Cohen, 2012), implying a decoupling between emotion and behavior. Heerey and Gold (2007) developed a computerized anticipatory and consummatory pleasure (ACP) task and found poorer emotion–behavior coupling in participants with chronic SCZ compared with healthy controls (HC), especially when they responded to internal rather than evoked stimulus representations. Using the same paradigm, this decoupling was also found in participants with first-episode (Lui, Liu, et al., 2016) and early stage SCZ (Lui, Shi, et al., 2016), as well as their unaffected first-degree relatives and schizotypal individuals (Xie et al., 2017). These lines of evidence suggest that emotion–behavior decoupling may be a trait marker for SCZ. However, emerging evidence from disorders such as MDD and BD also suggest the possibility of symptom-specificity but not diagnostic-specificity with respect to this deficit.

Indeed, a number of studies comparing individuals with MDD to HC have revealed reduced reactivity to positively valenced stimuli (Bylsma, Morris, & Rottenberg, 2008; Dunn, Dalgleish, Lawrence, Cusack, & Ogilvie, 2004). However, other studies have failed to observe such differences (Mitterschiffthaler et al., 2003; Renneberg, Heyn, Gebhard, & Bachmann, 2005; Sherdell, Waugh, & Gotlib, 2012) or have reported more nuanced patterns of emotion–behavior decoupling. For example, individuals with a current episode MDD, or subsyndromal individuals with depressive symptoms, but not remitted depression have all been shown to exhibit a lack of motivation to expend effort to pursue available rewards (Sherdell et al., 2012; Treadway & Zald, 2011; Yang et al., 2014). Furthermore, this deficit in motivation was also revealed to be differentially associated with symptoms of depression (Treadway & Zald, 2011; Yang et al., 2014). Together, this work suggests that emotion–behavior decoupling is likely to be a feature of MDD and associated with depressive symptoms, but to understand the role it plays in both illness etiology and treatment outcomes requires further study.

In BD (both Type I and II), several studies have reported anhedonia and avolition symptoms similar to those occurring in individuals with MDD (Gao, Sweet, Su, & Calabrese, 2017; Mula et al., 2010; Souery et al., 2012) and outpatients with SCZ (Strauss, Vertinski, Vogel, Ringdahl, & Allen, 2016). The reduced hedonic capacity has been found even in euthymic BD individuals (Di Nicola et al., 2013) and appears central to negative symptom networks (Strauss et al., 2019). Although these findings have provided some evidence of deficits in pleasure experienced in BD individuals, other studies using the Behavioral Inhibition and Behavioral Activation Scales have found comparable reward sensitivity between euthymic BD individuals and HC (Caseras, Lawrence, Murphy, Wise, & Phillips, 2013), but higher motivational salience and reward-seeking in individuals with BD (Alloy et al., 2008; Alloy et al., 2012; Alloy, Olino, Freed, & Nusslock, 2016; Johnson, Edge, Holmes, & Carver, 2012). Furthermore, the higher reward-seeking behavior was found to be positively correlated with manic symptoms (Alloy et al., 2008). Thus, emotion–behavior decoupling may occur in BD, although to fully understand its implications for the illness, the field needs further research, particularly research that relies less on self-report and more on measurable behavior.

These lines of evidence converge to support the idea that the nature of anhedonia/avolition in SCZ, MDD, and BD could be a failure in translating emotional salience into motivated behavior (Kring & Barch, 2014) instead of merely reflecting deficits in pleasure experience or anticipation. However, most previous studies have focused on the assessment of current emotional experience or the motivational aspect of anhedonia, and few studies have examined emotion–behavior coupling in different mental disorders. Furthermore, recent theoretical understanding of anhedonia has highlighted not only reduced experience in positive emotion but also elevation in negative emotion (Strauss, Visser, Lee, & Gold, 2017). Preliminary studies have found increased ambivalence in SCZ individuals (Strauss et al., 2017; Trémeau et al.,

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2009) and a hypersensitivity to loss in MDD individuals (Manelis et al., 2016; Smoski et al., 2008). However, few studies have explored the relationship between negative emotional experience and avoidance behavior in different psychiatric disorders, despite the potential importance of these relationships for treatment outcome.

According to Strauss and Cohen (2017), although negative symptoms such as avolition are transdiagnostic constructs, there may be multiple pathways leading to the same common endpoint of motivational deficits. Previous studies have found significant emotion–behavior decoupling in SCZ spectrum using the ACP task, but whether there is also such a deficit in MDD or BD individuals and whether there are similarities or differences among these diagnostic entities remains unclear. As such, a comparison study recruiting samples with different psychiatric disorders offers a unique opportunity to examine the equifinality phenomenon (whether the same behavioral manifestation is due to different neural mechanisms). Furthermore, comorbidity is common among mental disorders and about 45% of individuals with one mental disorder also meet diagnostic criteria for another (Kessler, Chiu, Demler, & Walters, 2005). In this context, the Research Domain Criteria framework (Insel et al., 2010; Miller, 2010) encourages the adoption of a dimensional approach in studying psychological constructs to circumvent the problems of diagnostic and symptom heterogeneity in psychiatric disorders. Although previous studies have suggested potential relationships between reward-seeking motivation with clinical symptoms (Alloy et al., 2008; Treadway & Zald, 2011; Yang et al., 2014), whether there are significant correlations between emotion–behavior decoupling with clinical symptoms among different mental disorders has not been tested.

In this study, we aimed to compare the phenomenon of emotion–behavior decoupling in participants with SCZ, MDD, and BD, using the well-validated ACP task, and explore the clinical correlates of this deficit across these clinical groups. We hypothesized that participants with SCZ and HC participants would rate positive and negative slides similarly in valence but SCZ would exhibit less motivation to seek positive slides and avoid negative slides, thus manifesting as a lower correspondence between emotion and behavior compared with HC. Furthermore, this decoupling would be observed in the representational responding phase according to the findings of Heerey and Gold (2007). We also hypothesized that emotion–behavior correspondence in participants with SCZ would be most severe among the three diagnostic groups. In participants with MDD, both similar and weaker responses to positive and negative stimuli than HC have been reported (Bylsma et al., 2008), but findings in motivational deficits are consistent (Sherdell et al., 2012; Treadway & Zald, 2011; Yang et al., 2014). Furthermore, most studies have reported deficits in both anticipatory and consummatory pleasure experience in individuals with MDD (Li et al., 2015; Yang et al., 2014). Based on these findings, we expected a lower emotion–behavior correspondence in individuals with MDD compared with HC. Although people with BD have been found to have reward hypersensitivity in most previous studies (Alloy et al., 2016), the association between slide valence rating and motivational behavior may also be abnormal. Given the available literatures, we hypothesized that individuals with BD when in a low and/or euthymic mood state would show decoupling, such that positive affect would be separated from motivated behavior. In addition, we also expected significantly positive correlations between negative symptoms and anhedonia with emotion–behavior decoupling across three clinical samples according to previous studies in each group (Gold et al., 2012, 2013; Strauss et al., 2019; Treadway & Zald, 2011).

Method

Participants

Based on effect sizes from previous studies (Heerey & Gold, 2007; Lui, Liu, et al., 2016), the sample size was estimated using G*power 3.1 software (Faul, Erdfelder, Lang, & Buchner, 2007). The software suggested a total sample size of 108 (effect size f = 0.25, α = .05, expected power = 0.8). Based on this estimation, 41 individuals with SCZ (39 inpatients, two outpatients), 44 individuals with MDD (30 inpatients, 12 outpatients), 43 individuals with BD (41 inpatients, two outpatients) and 43 healthy individuals were recruited. All clinical participants were recruited from the Mental Health Centre of Weifang City in China. Diagnosis was made in accordance with *DSM–IV* criteria and ascertained by the Mini-International Neuropsychiatric Interview (MINI) Version 6.0.0. Exclusion criteria included a lifetime history of alcohol and/or substance abuse, a history of neurological disorder or head injury, and an estimated IQ of less than 80.

Forty participants with SCZ were taking antipsychotics, including 14 participants taking one antipsychotic medication, 25 participants taking two antipsychotics, and one participant taking three antipsychotics. Among all participants with SCZ, 14 were taking mood stabilizers, two were taking antidepressants, and six were taking anxiolytics in addition to antipsychotics. Thirty-nine participants with MDD were taking antidepressants while five were medication-free. Fourteen patients with MDD were also taking antipsychotics, nine were taking mood stabilizers, and 13 were taking anxiolytics in addition to antidepressants. All but two participants (outpatients) with BD were taking medication. Fourteen were taking antipsychotic monotherapy, two were taking mood-stabilizer monotherapy, two were taking anxiolytics, and 15 were taking multiple medications. The medication details of eight participants were missing (for details, please see Supplementary Tables S1 and S2 in the online supplementary material). All participants were relatively clinically stable and capable of providing informed consent and completing experiments.

All healthy participants were recruited from the neighboring community. The MINI was also administered to all healthy controls by experienced psychiatrists (Guo-hui Zhu and Gui-zhen Wang) to ascertain the absence of any psychiatric disorders. This study was approved by the Ethics Committees of Weifang Medical University (Protocol 2016-008) and the Institute of Psychology, the Chinese Academy of Sciences (Protocol H15031). All participants provided written informed consent.

Clinical and Neuropsychological Assessments

The Positive and Negative Syndrome Scale (PANSS; Kay, Fiszbein, & Opler, 1987) and the Scale for the Assessment of Negative Symptoms (SANS; Andreasen, 1989) were used to assess the severity of schizophrenia symptoms. The 17-item Hamilton Rating Scale for Depression (HRSD-17; Hamilton, 1960) was used to assess the severity of depressive symptoms. The Mood Disorder

Questionnaire (MDQ; Hirschfeld et al., 2000) was used to assess the severity of manic symptoms. Intellectual functioning was estimated based on the short form (information, arithmetic, similarities and digit span subscales) of the Chinese version of the Wechsler Adult Intelligence Scale—Revised (WAIS–R; Gong, 1992). Working memory was assessed using the Chinese version of the Letter–Number Span Test (LNST; Chan et al., 2008), in which a series of alternating "Celestial Stems (an ancient Chinese system of ordinals) and Numbers" were read to participants. Participants were required to update and rearrange them in the correct order. The sum of all correct responses was used as the index of working memory ability. All clinical and neuropsychological assessments were conducted by trained psychiatrists and graduate students. The intraclass correlation coefficients of HRSD-17, the positive and the negative subscale of the PANSS, and the SANS were all greater than 0.80, showing a good reliability (please see Supplementary Table S3 in the online supplementary material).

Self-Report Instruments for Anhedonia

Three self-report instruments were selected to assess different facets of anhedonia. The Chinese version of the Temporal Experience of Pleasure Scale (TEPS; including the abstract anticipatory, the contextual anticipatory, the abstract consummatory and the contextual consummatory dimensions) was used to measure the anticipatory and consummatory facets of anhedonia (Chan et al., 2012; Gard, Gard, Kring, & John, 2006). The Chapman Physical Anhedonia Scale (CPAS) and the Revised Social Anhedonia Scale (RCSAS; Chapman, Chapman, & Raulin, 1976; Eckblad, Chapman, Chapman, & Mishlove, 1982) were used to assess the capacity to experience physical and interpersonal pleasure. The Anticipatory and Consummatory Interpersonal Pleasure Scale (ACIPS) was used to measure the capacity to anticipate and experience social interactions with other people (Gooding & Pflum, 2014). Higher scores on the TEPS and the ACIPS indicate better ability in experiencing anticipatory or consummatory pleasure, while higher scores on the CPAS and the RCSAS indicate impaired ability to experience pleasure.

The Computerized ACP Task

The ACP task developed by Heerey and Gold (2007) was used to evaluate emotion–behavior decoupling under representational and evoked responding conditions. The details of the task has been described elsewhere (Heerey & Gold, 2007; Lui, Liu, et al., 2016; Lui, Shi, et al., 2016; Xie et al., 2017). In brief, the task consisted of two phases in which participants made effortful key presses in response to images with different valences when they were off-screen (representational responding) and on-screen (evoked responding). During the first phase, 42 slides including 14 positive, 14 neutral and 14 negative slides were presented on the screen, each of which consisted of three pictures selected from the International Affective Picture System (IAPS; Lang, 2005) with equal valence and arousal ratings. These slides have been validated in the Chinese settings for SCZ individuals, unaffected first-degree relatives of individuals with schizophrenia, individuals with social anhedonia and healthy individuals (Lui, Liu, et al., 2016; Lui, Shi, et al., 2016; Xie et al., 2017). Participants were required to report how pleasant they felt and how aroused they were using nine-point Likert scales (from extremely unpleasant/calm to extremely pleasant/arousing) in response to these slides, and then press the buttons "m" and "n" or "x" and "z" alternately as quickly as possible with the anticipation to change the probability of stimulus exposure in the next phase (assessing anticipatory pleasure-seeking motivation and aversion-avoidance motivation). The button-pressing response window lasted for two seconds. During the evoked responding stage, 30 of the slides the participants viewed in the first phase were presented on the screen again. Participants were required to press the buttons "m" and "n" rapidly with the motivation to increase the exposure time or press the buttons "x" and "z" rapidly to shorten it (assessing consummatory pleasure-seeking motivation and aversion-avoidance motivation). The button-pressing response window for each trial varied from 2 to 10 s. If no button was pressed, a slide was shown for 5 s.

The valence and arousal ratings of each slide in the first phase, the mean button-pressing speed in each trial during the representational and the evoked responding phase were recorded.

Data Analysis

Data obtained from the behavioral paradigm were analyzed using the same method as described in previous studies (Heerey & Gold, 2007; Lui, Liu, et al., 2016; Xie et al., 2017). Briefly, four key parameters were obtained from the data, that is, self-reported liking and arousal, motivated behavior, and correspondence between liking and behavior. To analyze motivational salience under different emotional valences, two steps were carried out. First, slide valence ratings of 1–3, 4–6, and 7–9 were transformed into negative, neutral, and positive valences respectively on a participant-by-participant basis, as is standard for the task (Heerey & Gold, 2007). This ensured that slides were categorized according to each participant's own emotional experience, rather than standard slide ratings, thereby providing a more nuanced picture of affect behavior coupling.

Except for the neutral valence condition, if the valence rating was incongruent with button-pressing behavior, the trial was deemed invalid. For example, if a participant engaged in "avoidance" key-pressing to a slide they had rated positively, it would be difficult to determine whether the participant was doing so out of true motivation to avoid the image or out of confusion about which set of keys to press. Then, the button-pressing speed (presses per second) in each trial under different valence conditions was averaged and the index of motivational salience during the whole representational or evoked responding window was calculated accordingly. To avoid the influence of psychomotor slowing in the clinical groups, participants' button-pressing speeds were mean-centered (based on the participants' own mean button-pressing speed during all conditions; Xie et al., 2017). This score was used to conduct group comparisons on motivational salience. Regarding the correspondence between self-reported liking and behavior, correlation analyses between pleasantness/unpleasantness ratings and button-pressing speed to seek/avoid the stimuli were conducted and the resulting coefficients were transformed using Fisher's r to z transformation, prior to further analysis.

Statistical analysis was conducted using the Statistical Package for the Social Sciences (SPSS) Version 17.0. Descriptive analyses were conducted to describe the participants' sociodemographic characteristics. Pearson chi-square test was used for group comparisons for categorical variables. Analysis of variance (ANOVA)

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and post hoc tests with Bonferroni correction were conducted to examine group differences for continuous variables. Mixed-model ANOVAs were conducted to examine the between-subjects and within-subject effects of these key parameters from the behavioral data. Post hoc pairwise comparisons and subsequent simple effect analyses were conducted when significant main effects and interactions were detected. Pearson correlation analyses were used to calculate the correspondence coefficients and analyze the relationships between symptoms with task-based emotion–behavior coupling.

Results

Comparison of Demographic and Clinical Characteristics

As shown in Table 1, the four groups of participants were matched in gender and age ($ps > 0.05$). The SCZ, BD, and MDD groups had lower estimated IQ, $F_{(3,170)} = 15.558$, $p < .001$, poorer working memory, $F_{(3,169)} = 15.859$, $p < .001$, and more severe physical, $F_{(3,170)} = 9.995$, $p < .001$, and social anhedonia, $F_{(3,170)} = 16.015$, $p < .001$, compared with HC ($ps < 0.05$). As expected, the SCZ group showed the most severe impairment in working memory. The SCZ group also reported significantly lower abstract anticipatory pleasure on the TEPS than HC ($p < .05$).

Emotion–Behavior Decoupling as Measured by the ACP Task

Pleasantness ratings. The valence and arousal ratings of the four groups of participants are shown in Figure 1a. Regarding valence ratings, the main effect of valence, $F_{(2,166)} = 975.941$, $p < .001$, $\eta_p^2 = 0.854$, and the Group × Valence interaction, $F_{(6,334)} = 4.588$, $p < .001$, $\eta_p^2 = 0.076$, were both significant, but the main effect of group, $F_{(3,167)} = 1.249$, $p = .294$, $\eta_p^2 = 0.022$, failed to reach statistical significance. Further simple effect tests showed that participants with SCZ, MDD and BD all rated negative slides

Table 1
Comparisons of Demographic and Clinical Characteristics and Self-Reported Anhedonia Among Participants With Schizophrenia, Major Depressive Disorder, Bipolar Disorder, and Healthy Participants

Characteristic	SCZ (n = 41)		MDD (n = 44)		BD (n = 43)		HC (n = 43)		F/χ^2	p	Pairwise comparison (Bonferroni corrected)
	M	SD	M	SD	M	SD	M	SD			
Age (year)	27.32	6.81	28.66	8.80	30.88	8.26	27.93	5.53	1.842	.141	
Gender (male:female)	18:23		15:29		26:17		19:24		6.262	.100	
Education (year)	10.49	2.87	11.52	3.36	12.26	2.97	15.40	2.53	21.838	<.001	SCZ < BD*, SCZ < HC***, BD < HC***, MDD < HC***
Estimated IQ	93.10	13.47	99.70	16.00	99.51	17.80	116.30	17.77	15.558	<.001	SCZ < HC***, BD < HC***, MDD < HC***
Duration of illness (year)	2.63	2.62	2.91	3.82	6.15	5.55			8.940	<.001	SCZ < BD**, MDD < BD**
PANSS											
Positive scale	17.41	7.11	9.11	2.75	13.44	6.63			21.833	<.001	SCZ > BD > MDD ($ps < .01$)
Negative scale	17.02	7.54	11.70	5.67	13.00	6.72			7.283	.001	SCZ > BD*, SCZ > MDD**
General psychopathology scale	36.12	10.58	31.00	9.91	32.05	14.90			2.132	.123	
SANS	29.76	20.03	18.77	15.20	18.47	17.33			5.598	.005	SCZ > BD*, SCZ > MDD*
HRSD-17	13.44	8.02	15.11	8.10	13.65	8.51			.532	.589	
MDQ	6.32	3.37	4.39	3.04	7.51	3.69	6.86	4.47	5.841	.001	MDD < HC*, MDD < BD**
Working memory	8.56	4.64	11.98	5.26	11.56	4.35	15.49	4.16	15.859	<.001	SCZ < MDD**,BD*,HC***, MDD < HC**, BD < HC**
TEPS											
Abstract anticipatory	17.07	4.41	18.02	4.41	17.21	4.69	19.42	2.58	2.927	.035	SCZ < HC*
Contextual anticipatory	17.83	5.47	15.26	4.98	16.28	5.66	16.47	4.04	1.822	.145	
Abstract consummatory	25.54	5.92	26.00	6.03	27.09	7.02	27.37	4.12	.937	.424	
Contextual consummatory	15.41	4.14	15.51	4.05	15.49	4.73	16.19	3.70	.317	.813	
Chapman scale											
RCSAS	13.61	4.83	13.68	7.12	16.53	7.37	7.81	3.84	16.015	<.001	SCZ > HC***, BD > HC***, MDD > HC***
CPAS	21.27	8.60	21.41	9.63	20.84	10.00	12.60	7.00	9.995	<.001	SCZ > HC***, BD > HC***, MDD > HC***
ACIPS											
Anticipatory	30.12	7.32	30.45	5.68	29.77	8.18	32.98	4.45	2.123	.099	
Consummatory	42.22	11.39	42.50	8.38	42.19	13.09	47.09	6.70	2.384	.071	

Note. SCZ = schizophrenia; MDD = major depressive disorder; BD = bipolar disorder; HC = healthy controls; IQ = intelligence quotient; PANSS = Positive and Negative Syndrome Scale; SANS = the total score of the Scale for the Assessment of Negative Symptoms; HRSD-17 = the total score of the 17-item Hamilton Rating Scale for Depression; MDQ = total score of the Mood Disorder Questionnaire; TEPS = Chinese version of the Temporal Experience of Pleasure Scale; CPAS = Chapman Physical Anhedonia Scale; RCSAS = Revised Chapman Social Anhedonia Scale; ACIPS = Anticipatory and Consummatory Interpersonal Pleasure Scale.
* $p < .05$. ** $p < .01$. *** $p < .001$.

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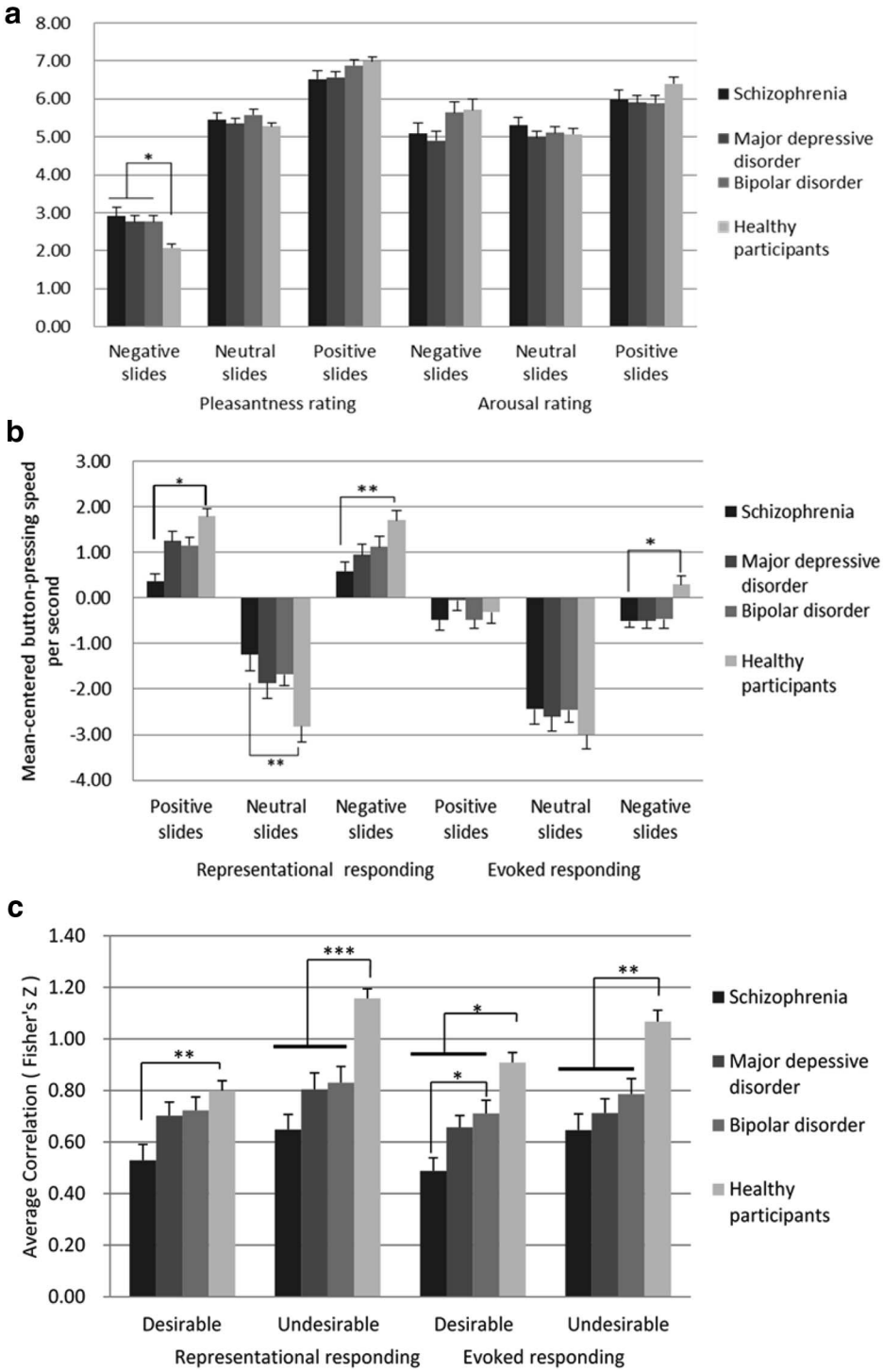

Figure 1. Comparisons of task performances. a: Pleasantness and arousal ratings for participants with schizophrenia, major depressive disorder, bipolar disorder, and healthy participants. b: Mean-centered button-pressing speed across slide valence and behavior conditions for participants with schizophrenia, major depressive disorder, bipolar disorder, and healthy participants. Participants' button-pressing speeds were mean centered based on the participants' own mean button-pressing speed during all conditions. c: Correspondence between motivated behavior and liking across slide desirability and condition in participants with schizophrenia, major depressive disorder, bipolar disorder and healthy participants. Data shows Fisher's r to Z transformation. Standard errors are represented in the figure by the error bars attached to each column. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

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more positively ($ps < 0.05$) but had similar valence ratings to neutral and positive slides ($ps > 0.1$) compared with HC.

Regarding arousal ratings, the main effect of valence was significant, $F_{(2,166)} = 28.874$, $p < .001$, $\eta_p^2 = 0.147$, and the pairwise comparisons revealed higher overall arousal ratings for positive stimuli than negative and neutral stimuli ($ps < 0.001$). However, the main effect of group, $F_{(3,167)} = 1.316$, $p = .271$, $\eta_p^2 = 0.023$, and the Group- × Valence interaction, $F_{(6,334)} = 1.167$, $p = .324$, $\eta_p^2 = 0.021$, both failed to reach statistical significance.

Motivational salience. Comparisons of the total number of button presses and average button-pressing speed in the ACP task among participants with SCZ, MDD, BD, and HC are shown in Table 2. Participants in all three clinical groups made fewer button presses and had slower average button-pressing speed than HC both in representational responding and evoked responding. Because of this psychomotor slowing, further group comparisons used mean-centered button-pressing speed data.

Figure 1b shows the mean-centered button-pressing speed to slides of different valences for participants with SCZ, BD, MDD, and HC. As expected, the main effect of group was not significant, $F_{(3,141)} = 0.648$, $p = .585$, $\eta_p^2 = 0.014$, suggesting a similar mean-centered button-pressing speed between the four groups. The predicted Group × Behavioral Condition × Valence interaction, $F_{(6,282)} = 3.539$, $p = .016$, $\eta_p^2 = 0.070$, and the Group × Valence interaction, $F_{(6,282)} = 6.442$, $p < .001$, $\eta_p^2 = 0.121$, were both significant. However, the expected interaction of Group × Behavioral Condition failed to reach statistical significance, $F_{(3,141)} = 0.266$, $p = .850$, $\eta_p^2 = 0.006$, suggesting that motivational behavior did not differ between representational responding and evoked responding among the four groups. Therefore, further mixed-model ANOVAs were conducted to analyze the Group × Valence interaction under each behavioral condition.

During representational responding, we found a significant Group × Valence interaction, $F_{(6,282)} = 6.967$, $p < .001$, $\eta_p^2 = 0.129$. Further simple effects tests showed that SCZ participants pressed buttons at the lowest speed among the four participant groups when they responded to positive slides ($ps < 0.05$) and pressed buttons at a lower speed than HC when they responded to neutral ($p = .005$) and negative slides ($p = .005$). However, contrary to expectations, there were no significant differences between BD and HC participants or between MDD and HC participants regardless of slide valence ($ps > 0.05$). During evoked responding, the Group × Valence interaction, $F_{(6,282)} = 3.270$, $p = .004$, $\eta_p^2 = 0.065$, was also significant. Further simple effects tests showed that participants from all three clinical groups pressed buttons at a significantly lower speed than HC when they responded to slides they had rated as negative ($ps < 0.05$). The difference between any two groups was not significant when they responded to positive ($ps > 0.05$) and neutral slides ($ps > 0.05$).

Correspondence Between Pleasantness Rating and Behavior

Figure 1c shows the correspondence (after Fisher's r to Z transformation) between button-pressing speed and pleasantness ratings. As expected, the mixed-model ANOVAs revealed a significant main effect for group, $F_{(1,167)} = 13.434$, $p < .001$, $\eta_p^2 = 0.194$, and all three clinical groups showed lower emotion–behavior correspondence compared with HC ($ps < 0.01$). Among the three clinical groups, participants with SCZ showed the worst correspondence compared with participants with BD ($p = 0.032$), whose correspondence was not significantly different from the MDD group ($p = 0.197$). The predicted Group × Valence interaction, $F_{(3,167)} = 6.063$, $p = .001$, $\eta_p^2 = 0.098$, and the Group × Behavioral Condition × Valence interaction, $F_{(3,167)} = 4.065$, $p = .008$, $\eta_p^2 = 0.068$, were both statistically significant. However, the expected Group × Behavioral Condition interaction, $F_{(1,167)} = 0.649$, $p = .585$, $\eta_p^2 = 0.012$, failed to reach statistical significance. Therefore, further mixed-model ANOVAs were conducted to analyze the Group × Valence interaction under each behavior condition.

During representational responding, simple effects tests showed that the correspondence between emotion and behavior in response to positive slides in SCZ participants was lower than that of HC ($p = .002$), while this decoupling was not found in MDD and BD participants ($ps > 0.1$). The correspondence between emotion and behavior in response to negative slides in all three clinical groups was significantly lower than that of HC ($ps < 0.001$). During

Table 2
Comparisons of Button Presses in ACP Task Among Participants With Schizophrenia, Major Depressive Disorder, Bipolar Disorder and Healthy Participants

Task performance	SCZ (n = 41)		MDD (n = 44)		BD (n = 43)		HC (n = 43)		F	p	Pairwise comparison (Bonferroni corrected)
	M	SD	M	SD	M	SD	M	SD			
Total number of button presses											
Representational responding	256.02	164.15	309.25	187.03	258.84	147.49	397.86	167.65	6.683	<.001	SCZ < HC** BD < HC**
Evoked responding	407.22	320.36	498.57	368.00	373.72	241.40	631.47	311.47	5.767	.001	SCZ < HC** BD < HC**
Average button-pressing speed											
Representational responding	6.65	3.98	7.69	4.57	6.72	3.71	9.69	4.18	5.046	.002	SCZ < HC** BD < HC**
Evoked responding	15.72	12.41	17.72	12.81	13.70	8.66	21.95	11.40	4.079	.008	SCZ < HC** BD < HC**
Invalid trials											
Representational responding	4.12	4.42	1.80	2.35	3.07	5.05	.79	1.44	6.857	<.001	SCZ > MDD* SCZ > HC*** BD > HC*
Evoked responding	3.73	3.46	1.82	2.72	2.81	3.89	.91	1.49	6.893	<.001	SCZ > MDD* SCZ > HC*** BD > HC*

Note. SCZ = schizophrenia; MDD = major depressive disorder; BD = bipolar disorder; HC = healthy controls.
* $p < .05$. ** $p < .01$. *** $p < .001$.

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evoked responding, the correspondence between emotion and behavior in response to both positive and negative slides in all three clinical groups were significantly lower than that of HC ($ps < 0.05$). Furthermore, SCZ participants also showed more severe emotion–behavior decoupling in response to positive slides than BD participants ($p < .05$).

Correlation Analysis

The results of correlation analyses between symptoms with emotion–behavior correspondence in the three diagnostic groups are shown in Table 3. Pearson correlation analyses revealed negative correlation between physical anhedonia and emotion–behavior correspondence under each condition, and the coefficient of the association between physical anhedonia and correspondence in the undesirable condition during evoked responding survived Bonferroni correction, $r = .353$, $p < .05$. Consistent with our hypothesis, all emotion–behavior correspondences except for the desirable condition during evoked responding were negatively correlated with negative symptoms. The correlation between emotion–behavior correspondence in the desirable condition during representational responding and negative symptoms also survived after Bonferroni correction, $r = -0.275$, $p < .05$.

Discussion

The present study examined the similarities and differences in emotion–behavior decoupling between participants with SCZ, BD, and MDD, using the same set of self-report scales and behavioral paradigm. Consistent with our hypothesis, all three clinical groups showed lower emotion–behavior correspondence compared with HC, implying a common emotion–behavior decoupling among participants with SCZ, MDD and BD. Compared with BD and MDD participants, participants with SCZ showed the lowest emotion–behavior correspondence, in which both positive and negative images elicited less motivated behavior in both representational and evoked responding. In MDD and BD participants, emotion–behavior decoupling to avoid negative stimuli was observed in representational and evoked responding, whereas emotion–behavior decoupling in pleasure seeking was only observed in evoked responding (see Figure 2). In the combined sample of SCZ, MDD, and BD participants, several indices of emotion–behavior decoupling showed significant inverse correlations with negative symptoms and physical anhedonia. However, the magnitude of these correlations was generally small and only two survived correction for multiple comparisons.

Table 3
Correlations Between Symptoms With Emotion–Behavior Correspondence (r)

Measure	Representational desirable	Representational undesirable	Evoked desirable	Evoked undesirable
RCSAS	−.086	−.011	−.104	−.192*
CPAS	−.229**	−.209*	−.241**	−.353**[a]
HRSD-17	.032	.067	.093	.096
PANSS_P	.012	.071	−.048	.050
PANSS_N	−.275**[a]	−.239**	−.137	−.248**
MDQ	.130	.138	.043	−.001

Note. CPAS = Chapman Physical Anhedonia Scale; RCSAS = Revised Chapman Social Anhedonia Scale; PANSS_P = Positive Syndrome Scale; PANSS_N = Negative Syndrome Scale; HRSD-17 = total score of the 17-item Hamilton Rating Scale for Depression; MDQ = the total score of the Mood Disorder Questionnaire.
[a] Significant after Bonfferoni correction.
* $p < .05$. ** $p < .01$.

			SCZ	MDD	BD
Valence rating		Positive			
		Neutral			
		Negative	↑HC	↑HC	↑HC
Arousal rating		Positive			
		Neutral			
		Negative			
Button-pressing speed	Representational responding	Positive	↓↓↓MDD, BD & HC		
		Neutral	↓HC		
		Negative	↓HC		
	Evoked responding	Positive			
		Neutral			
		Negative	↓HC	↓HC	↓HC
Emotion-behavior correspondence	Representational responding	Positive	↓HC		
		Negative	↓HC	↓HC	↓HC
	Evoked responding	Positive	↓↓BD & HC	↓HC	↓HC
		Negative	↓HC	↓HC	↓HC

Figure 2. Summary of results. The ↑ symbol indicates that the specified group had higher scores relative to healthy controls (HC). ↓ means lower scores compared to HC. ↓ ↓ indicates that the group scored lower than the bipolar disorder (BD) and HC groups. ↓ ↓ ↓ indicates the lowest scores among the four groups. Blank cells indicate no group difference. SCZ = schizophrenia; MDD = major depressive disorder.

To the best of our knowledge, this is the first study that examines the extent of emotion–behavior decoupling across several different diagnostic entities and directly compares self-reported and behavioral anhedonia and avolition in individuals with SCZ, MDD, and BD. We observed the most serious emotion–behavior decoupling in SCZ participants, not only in both responding phases (representational, evoked), but also in both kinds of valence conditions (positive, negative). Furthermore, the button-pressing speed in response to positive slides in SCZ participants was also the lowest among the three clinical groups. In the study by Heerey and Gold (2007), clinically stable outpatients with SCZ only showed emotion–behavior decoupling in representational responding. However, Lui, Liu, et al. (2016) found that participants with first-episode SCZ showed emotion–behavior decoupling in avoiding negative stimuli in both representational and evoked responding and showed decoupling in seeking pleasure in evoked responding only, which is similar to our findings in MDD and BD participants. Most of our participants with SCZ were inpatients undergoing treatment, whereas those in the study of Heerey and Gold (2007) were clinically stable outpatients. Furthermore, the severity of positive and negative symptoms in the SCZ participants in the study by Lui, Liu, et al. (2016) was lower than those in our study, but similar to participants with MDD and BD in our study. Our correlation analyses also revealed significantly negative correlations between negative symptoms with emotion–behavior decoupling across the three clinical samples. Therefore, the differences in the severity of clinical symptoms may partly explain the inconsistent results.

Another possible reason may be related to working memory function. Evidence has consistently supported that working mem-

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ory plays a pivotal role in the active maintenance of emotional information, value and effort computation, and the ability to use prior emotional experience to guide current behavior (Kring & Barch, 2014). A previous study has also suggested that better working memory predicts higher efficiency in translating emotion into motivated behavior (Heerey & Gold, 2007). In our previous study using a first-episode SCZ sample (Lui, Liu, et al., 2016), decoupling of emotion and aversion-avoidance behavior only occurred in the SCZ subgroup with poor working memory, but not in those with intact working memory. Compared with our earlier study, the SCZ participants in this study had poorer working memory (M = 8.56, SD = 4.64) than our earlier sample (M = 11.80, SD = 2.37). This difference in working memory impairment might explain the divergent findings between this and the earlier study. Moreover, working memory performance differed significantly among the clinical groups, with SCZ participants having the worst, BD participants having intermediate, and MDD participants having the mildest impairment in working memory. A supplemental analysis revealed significant positive correlations between emotion–behavior correspondence and working memory in the combined clinical sample (please see Supplementary Table S5 in the online supplementary material).This pattern corresponds well to the difference in emotion–behavior decoupling observed among the three clinical groups.

Consistent with our hypotheses, participants with MDD demonstrated emotion–behavior decoupling both in representational and evoked responding, especially in avoiding negative stimuli. Furthermore, participants with MDD were also found to expend less effort in button-pressing than HC only under aversion-avoidance conditions. Compared with SCZ participants, the severity of negative symptoms assessed by the PANSS in participants with MDD was lower. The relatively intact motivation for pleasure-seeking in participants with MDD may reflect a milder functional impairment than SCZ participants. As such, we speculate that aversion-avoidance may be impaired before pleasure-seeking behavior with the onset of illness. The decoupling difference between positive-valenced conditions (M = 0.831) and negative-valenced conditions (M = 0.691) among the four groups also provides preliminary evidence supporting this speculation. However, the underlying mechanism of this phenomenon warrants further investigation.

It is unexpected that BD participants exhibited a similar pattern of emotion–behavior decoupling as MDD participants. In most previous studies, patients with BD were found to have reward hypersensitivity and higher reward pursuit motivation, even when euthymic (Alloy et al., 2008, 2016). However, in our study, participants with BD showed a similar valence rating to positive stimuli, but a lower pleasure-seeking behavior compared with HC. Alloy et al. (2016) has proposed the Reward Sensitivity Model, which suggests that reward hypersensitivity is a mood-independent feature of BD. Hypersensitivity of the reward system results in overreaction toward cues signaling possible attainment and loss of rewards (Alloy et al., 2016), leading to mania/hypomania or bipolar depression respectively. Our BD participants had similar negative symptom and depression scores as MDD participants, which may explain the lower pleasure-seeking behavior observed. Moreover, the positive stimuli we used in the present study were emotional pictures, which may have different incentive effects as monetary rewards (Li et al., 2018). Even so, our findings still suggest that the nature of anhedonia in BD is more likely to be due to decoupling of emotion experience and motivational behavior instead of an unstable reward system.

It is interesting that participants with SCZ, MDD, and BD reported less unpleasant emotion to negative slides but comparable pleasure experience to positive slides compared with HC. Some previous studies have reported the phenomenon of "emotional paradox" in SCZ participants (Lui, Liu, et al., 2016; Strauss & Gold, 2012), which refers to the presence of significant anhedonia when assessed with self-report scales but relatively intact "in-the-moment" experiential pleasure in behavioral performance. Our results support the existence of this phenomenon. This may seem counterintuitive, but affective ambivalence (Strauss et al., 2017; Trémeau et al., 2009) and coactivation (Strauss et al., 2017) have been reported in some previous studies (Sanchez, Lavaysse, Starr, & Gard, 2014; Strauss & Herbener, 2011; Strauss, Visser, Keller, Gold, & Buchanan, 2018; Trémeau et al., 2009). Strauss et al. (2017) have demonstrated that individuals with SCZ experience greater "coactivation" of positive and negative emotions than HC, which may lead to a blurred affective state, and contribute to a failure in translating emotions into motivated behavior. Although negative bias and positive offset have been reported in MDD (Gollan et al., 2015; Gollan et al., 2016) and SCZ individuals (Strauss et al., 2017), whether or not coactivation of positive and negative emotions also exists in BD individuals requires further investigation.

We also found significant physical and social anhedonia in the three clinical groups, which is consistent with previous studies (Tso, Grove, & Taylor, 2014; Winer, Jordan, & Collins, 2019; Zou et al., 2018). However, scores on most of the facets of the TEPS and the ACIPS in the three clinical groups were not significantly different from HC, except for the abstract anticipatory pleasure in SCZ participants. Some previous studies have found a deficit in anticipatory pleasure but not consummatory pleasure experience using the TEPS in participants with SCZ (Chan et al., 2010; Favrod, Ernst, Giuliani, & Bonsack, 2009; Gard, Kring, Gard, Horan, & Green, 2007). Schlosser et al. (2014) even found no significant impairment in anticipatory and consummatory pleasure experience in individuals with early course or late-course SCZ. Similar to our results in BD participants, Tso et al. (2014) also found significant physical and social anhedonia using the RCPAS and the RCSAS, but intact anticipatory and consummatory pleasure experience using the TEPS in BD participants. The RCPAS and the RCSAS are mainly used to assess trait anhedonia, whereas the TEPS and the ACIPS are more sensitive to the current illness state (Rizvi, Pizzagalli, Sproule, & Kennedy, 2016). Most of our clinical participants were under medication treatment, which may be one possible reason for the inconsistent results. In addition, research has shown that age correlates with neural processing of reward anticipation (Urošević, Luciana, Jensen, Youngstrom, & Thomas, 2016) and the incidence of anhedonia (Lampe, Kahn, & Heeren, 2001). Most of our clinical participants were in their early adulthood and the early stages of illness, which may be another reason for their relatively intact anticipatory and consummatory pleasure experience.

This study has several limitations. First, we did not classify our BD participants into different subtypes and different stages of illness. The heterogeneity of diagnosis, variation in symptom severity and differences in the nature of the last preceding illness episode in this group might have affected the results. Second, the

clinical groups included a combination of inpatients and outpatients, and the ration of these subsamples were unbalanced across the three clinical samples, which might also have effect on the results (please see Supplementary Table S4 in the online supplementary material). Third, our results might have been confounded by the effects of medications in our clinical participants. In addition, we have focused on the valence effect of slides on motivation but not the arousal effect. There was therefore a lack of comparison in motivational salience and emotion–behavior correspondence across slides at different arousal levels. Lastly, observational variables under different behavioral conditions may have linear correlations. The mixed-model ANOVAs conducted in our study may to some extent violate assumptions of independence of observations. Therefore, future work should aim to recruit a sufficient sample of participants to allow the use of more complex multilevel models that include appropriate random effects.

Taken together, our findings suggest that individuals with SCZ, BD, and MDD each experience a degree of anhedonia and avolition. Task-based results revealed different patterns of emotion–behavior decoupling among participants with SCZ, BD, and MDD, such that both positive and negative images elicited less motivated behavior in both representational and evoked responding in participants with SCZ, whereas the deficits in emotion–behavior coupling in participants with MDD and BD mainly occurred in negative stimulus avoidance, as well as positive stimulus seeking during evoked responding. In the combined clinical sample, deficits in aspects of emotion–behavior correspondence showed some associations with negative symptoms and anhedonia. Our findings provide a potentially common psychopathological mechanism of anhedonia and avolition across different mental disorders, which may be helpful for early detection of these disorders and the development of novel interventions.

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